

NAME : ABD UR REHMAN

ID : 39378

TASK : LAB 05

TASK CODE:

1. [DDL] Add Email column:

ALTER TABLE Employee

ADD Email VARCHAR(100);

2. [DML] Insert new department:

INSERT INTO Department (DeptID, DeptName)

VALUES (60, 'R&D');

3. [DML] Update salary of Bob Johnson (EmpID 102):

UPDATE Employee

SET Salary = Salary * 1.10

WHERE EmpID = 102;

4. [DQL] Employees hired after 2020-01-01:

SELECT Name, Salary

FROM Employee

WHERE HireDate > '2020-01-01';

5. [DML/DDDL] Delete employee and drop column:

DELETE FROM Employee

WHERE EmpID = 103;

ALTER TABLE Employee

DROP COLUMN Email;

6. [FK/DML] Insert employee with invalid DeptID (99):

INSERT INTO Employee (EmpID, Name, DeptID)

VALUES (999, 'Test User', 99);

Expected Result:

✗ Error due to foreign key constraint violation

Reason: DeptID 99 does not exist in Department table

7. [FK/DML] Delete Sales department (DeptID 10):

DELETE FROM Department

WHERE DeptID = 10;

Expected Effect (ON DELETE SET NULL):

Employees in Sales will NOT be deleted

Their DeptID will become NULL

8. [FK/DQL] Employees with NULL DeptID:

```
SELECT *  
  
FROM Employee  
  
WHERE DeptID IS NULL;
```

9. [FK/DML] Update DeptID from 20 to 200 (CASCADE):

```
UPDATE Department  
  
SET DeptID = 200  
  
WHERE DeptID = 20;
```

Expected Effect (ON UPDATE CASCADE):

Department ID changes to 200

All employees with DeptID = 20 automatically updated to 200

10. [FK/DQL] Check WorksOn after deleting EmpID 103:

```
SELECT *  
  
FROM WorksOn;
```

Expected Effect (ON DELETE CASCADE):

Any rows in WorksOn related to EmpID 103 are automatically deleted

No orphan records remain for EmpID 103